

Yellow sheet: Cells

20 marks | Approximately 22 minutes

DRAFT PAPER FOR TEACHER REVIEW

Class: _____ Date: _____

Answer all six questions. Use scientific terms and give reasons where requested.

1. Cell structures [4]

Name the cell structure that performs each function.

a. Contains most of the cell's genetic information in a typical eukaryotic cell. [1]

b. Regulates the movement of substances into and out of a cell. [1]

c. Is the site of protein synthesis. [1]

d. Is the site of aerobic respiration in a typical plant or animal cell. [1]

2. Classify using evidence [6]

For each organism, state its group and give one supporting feature. Choose from plant, animal, fungus, protocist or bacterium.

a. Organism P consists of one cell. It has no membrane-bound nucleus and its circular chromosome lies in the cytoplasm. [2]

b. Organism Q has a nucleus and a cell wall made of chitin. It does not have chloroplasts. [2]

c. Organism R lives as a single cell in fresh water. It contains a nucleus and chloroplasts. [2]

Cells - continued

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3. A root cell [2]

A cell from a plant root has a cellulose cell wall but no chloroplasts. A pupil claims it must be an animal cell because it cannot photosynthesise. Explain why this claim is incorrect. [2]

4. Differentiation [2]

An unspecialised plant cell develops into a palisade cell containing many chloroplasts. Name this process and explain how the chloroplasts help the cell carry out its function. [2]

5. Stem cells [4]

a. State two key abilities of a stem cell. [2]

b. Compare the range of specialised cells that embryonic stem cells and adult tissue stem cells can produce. [2]

6. Applying stem-cell knowledge [2]

Researchers propose using stem cells to replace damaged specialised cells. Explain one potential benefit and one possible biological risk of this approach. [2]

Total: 20 marks